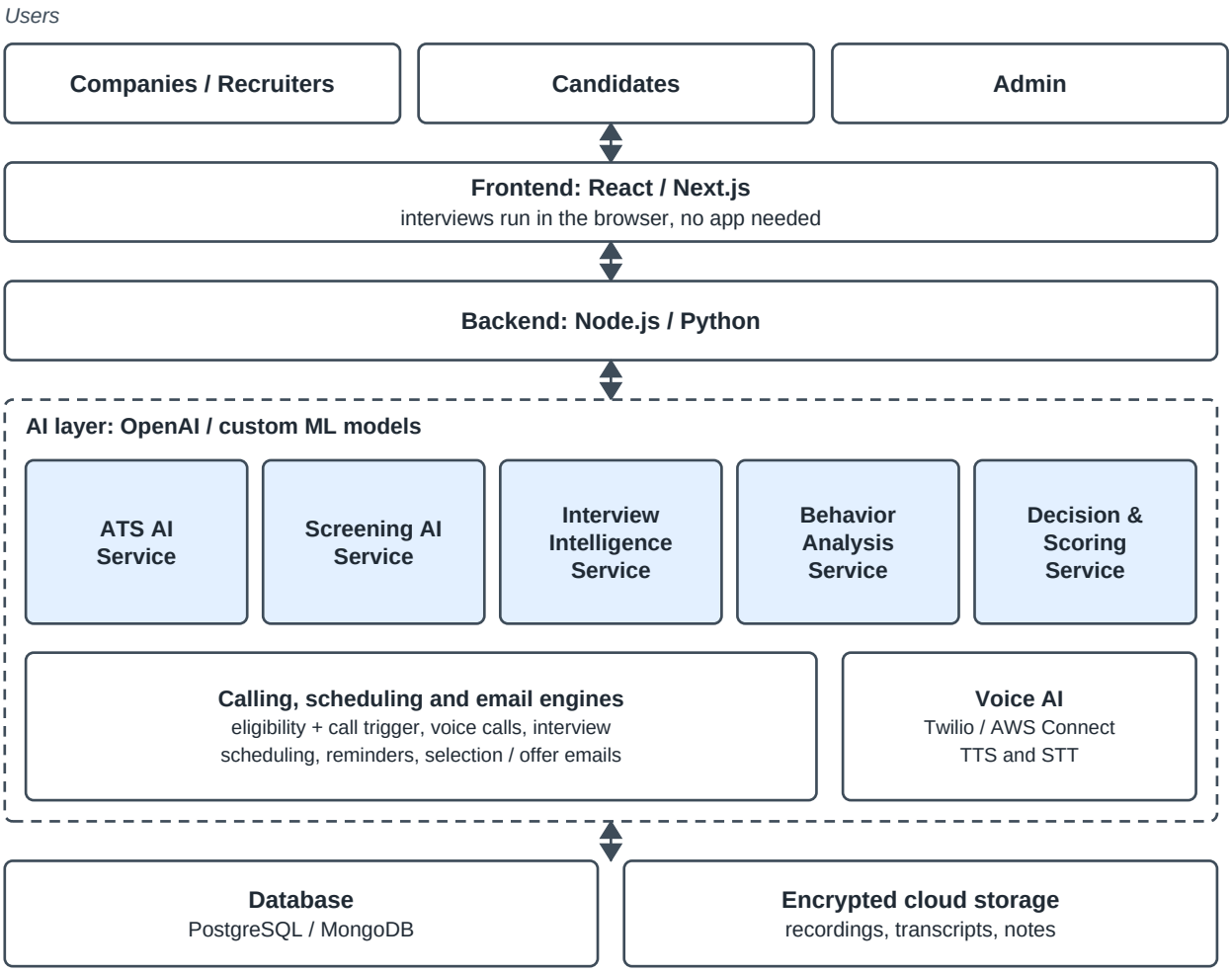
Zecpath AI System Architecture

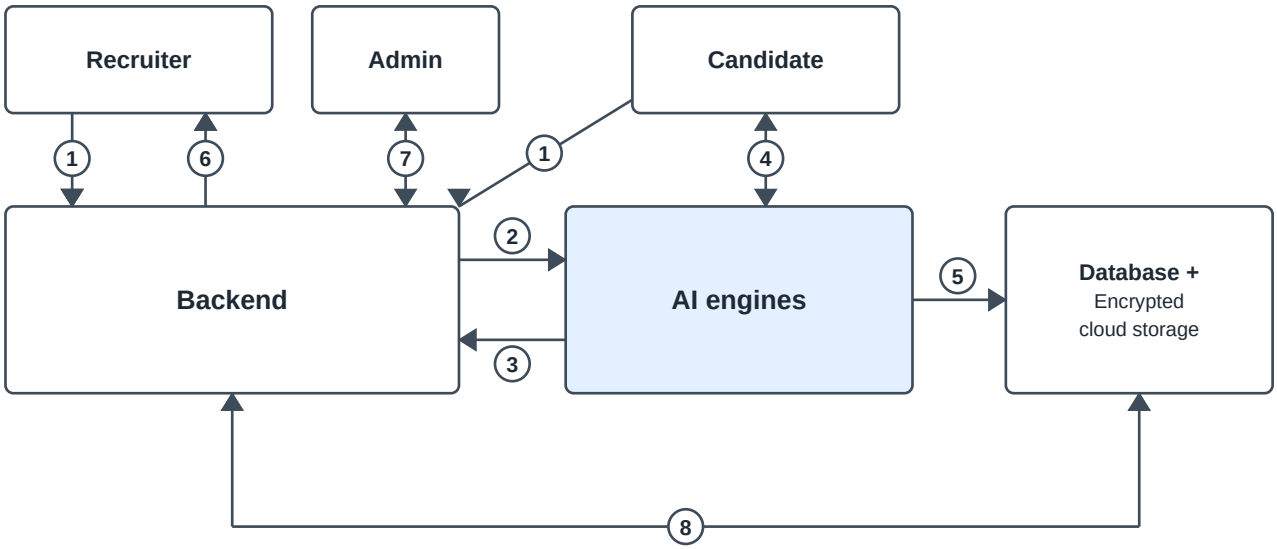
Day 2 task | Prepared by Abhishek | Based on the Zecpath PRD (Phases 1 to 100)

1. AI system architecture



The AI is split into 5 services, so each one does one job and can be changed on its own. Users only talk to the frontend. The backend sends work to the AI layer and saves the results. Tech names are from the suggested tech stack in the PRD.

2. Data flow (Backend, AI engines, Storage)



Numbers match the table below

#	Flow	What moves
1	Recruiter / Candidate to Backend	Job post and requirements, candidate profile, resume, consent for AI interview and recording
2	Backend to AI engines	Resumes, job requirements, shortlisted candidates
3	AI engines to Backend	Scores, call status, decisions, hiring fit %, summary
4	Candidate and AI engines	AI calls, spoken answers, browser video interview, reminder calls
5	AI engines to Storage	Video, audio transcript, notes, screen recording, scores
6	Backend to Recruiter	Dashboard with results (preview first, full details after payment)
7	Admin and Backend	User and company management, call logs, voice and language settings
8	Backend and Storage	Consent records, logs, stored recordings for playback

3. Input / output of each AI service

AI service	Receives (input)	What it does	Returns (output)
ATS AI Service	Resume (PDF, DOC, DOCX) and job requirements	Reads the resume, matches skills and experience with the job, scores and ranks	ATS score and shortlist for each job
Screening AI Service	Candidate's spoken answers to basic questions (experience, skills, notice period, expected salary)	Converts speech to text, checks and scores the answers	Screening report
Interview Intelligence Service	Candidate's answers in the video interview, plus job role and experience	Asks HR and technical questions with follow-ups, checks communication and aptitude, gives a coding / task test, handles final salary discussion	Communication score (0 to 100), aptitude score, technical score, machine test result, negotiation outcome
Behavior Analysis Service	Camera feed and browser / screen activity	Checks the face is visible and only one person is there, watches eye and head movement, tab switching and phone use	Warnings, malpractice flags, behavior score
Decision & Scoring Service	ATS, HR, technical, machine test and behavior scores, negotiation outcome	Combines the scores (weights can be set per role)	Decision (selected / hold / rejected / offer recommended), hiring fit %, AI summary

Other helper engines

These do not score candidates, but the system needs them: **calling and voice** (calls eligible candidates in office time, retries after 1 hour), **scheduling and reminders** (sets the interview time, sends the email and reminder call), **emails** (selection, rejection, offer letter) and **recording and storage** (keeps interview video and notes in encrypted cloud storage).